Theory Assignment Solutions

Problem 1: The Generalization of Eigen-Theory

Task A: Standard eigendecompositing requires the input and output vectors to exist in the exact same vector space so that they can be compared (x and Ax must be the same length). A 5000×10 matrix takes a 10-dimensional input vector and maps it to a 5000-dimensional output vector. You cannot define an eigenvector if the input and output spaces are entirely different.

Task B: For the full SVD, U is 5000×5000, Σ is 5000×10, and V^T is 10×10. (In reduced SVD, U becomes 5000×10 and Σ become 10×10.

Task C: An orthogonal matrix means its columns and rows are mutually perpendicular, unit vectors (orthonormal). A key property of orthogonal matrices is that their transpose is exactly equal to their inverse. Therefore, $V^{-1} = V^T$

Problem 2: The Geometry of SVD

Task A: V^T performs a pure rotation (or reflection) on the input vector x, aligning the with the true orthogonal axes of variance in the input space.

Task B: Σ performs a pure scaling (stretching or squishing) along those newly aligned axes the singular values dictate the magnitude of the strecth.

Task C: If a singular value is 0, the matrix crushes that entire dimension down to a point the output space loses a dimension (e.g., a 3D sphere flattens into a 2D disk) This directly correspond to vectors falling into the Null space, representing loss of

invertible information.

Problem 3: The PCA Connection

Task A: $X^T X = (U\Sigma V^T)^T (U\Sigma V^T)$

Task B:

$$(U\Sigma V^T)^T (U\Sigma V^T) = (V^T \Sigma^T U^T)(U\Sigma V^T)$$

$$= V\Sigma^T (U^T U)\Sigma V^T$$

Because U is orthogonal. $U^T U = I$. Because Σ is a diagonal, $\Sigma^T = \Sigma$.

$$= V\Sigma I \Sigma V^T$$

$$= V\Sigma^2 V^T$$

Task C: By comparing $X^T X = V\Sigma^2 V^T$ to standard eigendecomposition $C = QAQ^T$, we mathematically prove that the columns of V (the right singular vectors) are exactly the Principal components (eigenvectors) of the covariance matrix. Furthermore, the eigenvalues Λ are exactly equal to the squared singular values Σ^2.